Roberto Turibio **E. K. Martins** | Experimental Physicist

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Find me

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ResearchGate: researchgate.net/profile/Roberto-Turibio-Martins

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Education

University of São Paulo (USP) — **São Paulo, Brazil**
M.Sc. in Physics — *2026–current*
Advisor: Prof. Neilo M. Trindade. FAPESP fellow (#2025/26960-8).

Federal Institute of São Paulo (IFSP) — **São Paulo, Brazil**
B.Sc. in Physics (Teaching track) — *2021–2025*
Advisor: Prof. Neilo M. Trindade. FAPESP fellow (#2022/07200-4).

Research Experience

University of São Paulo (USP) — **São Paulo, Brazil**
Master's Research Fellow — *Mar. 2026 – Feb. 2028*
Phototransfer and thermally assisted mechanisms in potassium feldspars ($KAlSi_3O_8$). FAPESP #2025/26960-8.

Technical University of Denmark (DTU) — **Roskilde, Denmark**
Research Internship Abroad — *Apr. 2024*
Time-resolved optically stimulated luminescence of amazonite. FAPESP-BEPE #2024/00890-0. Supervisor: Prof. Mayank Jain.

University of São Paulo (USP) — **São Paulo, Brazil**
Undergraduate Research Fellow — *Sep. 2022 – Dec. 2025*
Optical and thermoluminescence properties of Brazilian amazonite. FAPESP-IC #2022/07200-4.

Federal Institute of São Paulo (IFSP) — **São Paulo, Brazil**
Undergraduate Research Scholar — *Sep. 2021 – Aug. 2022*
Thermoluminescence of minerals from southern Minas Gerais. PIBIC #119/2021.

Publications

[1] R. T. E. K. Martins, M. C. S. Nunes, L. Bossin, E. G. Yukihara, N. M. Trindade. *Luminescence emission bands of amazonite ($KAlSi_3O_8$) under different excitation modes.* Radiation Physics and Chemistry, v. 250, p. 114299, 2026.

[2] R. T. E. K. Martins, M. L. Chithambo, N. M. Trindade. *Effects of blue illumination on the thermoluminescence of amazonite ($KAlSi_3O_8$).* Physica B: Condensed Matter, v. 728, p. 418353, 2026.

[3] R. T. E. K. Martins, E. M. Yoshimura, N. M. Trindade. *Optically stimulated luminescence of amazonite using linear modulated technique.* Journal of Luminescence, v. 289, p. 121610, 2025.

[4] R. T. E. K. Martins, R. Künzel, C. Ulsen, E. M. Yoshimura, N. M. Trindade. *Structural, optical and thermoluminescence properties of amazonite.* Radiation Physics and Chemistry, v. 223, p. 111947, 2024.

[5] R. T. E. K. Martins, I. A. Ferreira, A. O. Silva, M. C. S. Nunes, C. Ulsen, R. Künzel, M. M. Souza, M. L. Chithambo, E. M. Yoshimura, N. M. Trindade. *Thermoluminescence of rose quartz from Minas Gerais, Brazil.* Radiation Physics and Chemistry, v. 209, p. 110960, 2023.

Awards

2023: Bernhard Gross Award — Brazilian Materials Research Society (SBPMat).

2023: Honorable Mention, Best Poster — Thematic Area "Minerals and Geological Materials", Autumn Meeting of the Brazilian Physical Society (SBF).

Teaching & Outreach

Federal Institute of São Paulo (IFSP) **Guarulhos, Brazil**
Pedagogical Residency Program (volunteer) *2023–2024*
ENEM preparation project at Escola Estadual Cyro Barreiros. Supervisors: Prof. W. G. Schmiedecke and Prof. D. A. Moura. Preceptor: R. S. Anunciação.
2023: Talk: *Undergraduate research in the Physics Teaching track at IFSP.*

2022–2023: Outreach articles: IFSP news *"Meu câmpus tem estudo de minerais para aplicação em dosimetria das radiações"* and IFUSP news on the SBPMat award.

Technical Skills

Programming: Python (NumPy, SciPy, Matplotlib, pandas), LabVIEW

Computational Methods: Non-linear fitting, numerical solution of ODEs/PDEs, Monte Carlo simulations

Scientific Software: LaTeX, X'Pert HighScore Plus, VESTA, UnitCell, PyMatGen, Linux, Git

Structural Characterization: XRD (including Rietveld refinement), XRF, FTIR, Raman

Optical & Luminescence: UV-Vis, TL, OSL, LM-OSL, TR-OSL, PL, RL

Languages

Portuguese: Native

English: C1 (advanced)

Spanish: B1 (intermediate)

Russian: Basic

Selected Courses

2026: V Course on Rietveld Method with Nimloth (16h), UNIFEI

2025: Ionizing Radiation Interaction with Solids (30h), USP

2024: Electronics Assistant (80h), SENAI

2024: Python Fundamentals I (30h), SENAI

2023: Fundamentals and New Developments in Luminescence Dosimetry, IFUSP

2023: Scientific Communication and Writing (4h), ACS